

Asif V M

Kerala , India | Ready to Relocate

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PROFESSIONAL SUMMARY

AI and Machine Learning Engineer with an M.Sc. in Computer Science (Data Science specialization) and proven experience delivering production-grade ML systems. Built an institutional-level algorithmic trading engine featuring Hidden Markov Model regime detection, Black-Litterman portfolio optimization, and CVaR risk management – fully deployed with FastAPI, Streamlit, and live Alpaca API integration. Experienced across the full ML lifecycle: data pipelines, model development, deployment, and monitoring. Proficient in Python, PyTorch, TensorFlow, HuggingFace, NLP, time-series forecasting, and AWS cloud infrastructure. Multilingual (English, Malayalam, Tamil, Hindi, Arabic) with strong communication skills. Seeking ML Engineer and Data Scientist roles in UAE, Saudi Arabia, and the wider GCC region – available immediately.

EDUCATION

M.Sc. Computer Science – Data Science & Big Data Analytics <i>Yenepoya University, Bengaluru, India</i>	July 2026 CGPA: 8.0/10
B.Sc. Computer Science with Data Science <i>Nehru Arts and Science College, Coimbatore, India</i>	2021 – 2024 CGPA: 8.1/10

PROFESSIONAL EXPERIENCE

Quantum Learnings <i>Data Science Intern</i>	Feb 2025 – Jun 2025 <i>Bengaluru, India</i>
- Delivered an institutional-grade algorithmic trading engine as the primary internship project, gaining first industry experience building production ML systems outside an academic environment - Engineered a quant trading system featuring Hidden Markov Model (HMM) market regime detection, Black-Litterman portfolio optimization, and CVaR-based risk management – equivalent in architecture to institutional trading desks - Built full-stack application with FastAPI backend and Streamlit dashboard, integrated with Alpaca paper trading API for live order execution and real-time portfolio monitoring across NSE and NYSE markets - Implemented dynamic position sizing, portfolio rebalancing logic, and multi-asset allocation with automated risk alerts – delivering a complete, deployable product within the internship period	

TECHNICAL SKILLS

Machine Learning & AI: Scikit-learn, PyTorch, TensorFlow, HuggingFace Transformers, HMM (hmmlearn), NLP, Sentiment Analysis, Time Series Forecasting, Anomaly Detection, Ensemble Methods, Fine-tuning LLMs
Quantitative & Statistical: Black-Litterman Optimization, CVaR Risk Modeling, Portfolio Theory, Hypothesis Testing, Feature Engineering, A/B Testing
Programming & Data: Python (Pandas, NumPy, SciPy), SQL (Advanced – Window Functions, CTEs), R (Basic)
Deployment & MLOps: FastAPI, Flask, Docker, Streamlit, MLflow (basic), AWS (S3, RDS, EC2), Azure, RESTful APIs, CI/CD pipelines
Visualization & BI: Power BI, Plotly, Matplotlib, Seaborn, Interactive Dashboards
Tools: Git/GitHub, Jupyter, MySQL, PostgreSQL, SQLite, ETL Pipelines, Alpaca Trading API

KEY ACHIEVEMENTS

- Built institutional-grade algorithmic trading engine with HMM regime detection achieving 85% directional accuracy on 7–30 day stock forecasts across NSE and NYSE markets
- Deployed 5+ full-stack ML applications with live web interfaces processing 100K+ records each, accessible via public Streamlit URLs
- Designed statistical outlier detection pipeline for 100K+ electoral records achieving 95% precision using Z-score and IQR analysis
- Completed 5 professional certifications spanning IBM Data Science, Google Cloud Analytics, and Microsoft Azure Fundamentals (AZ-900)
- Built and fine-tuned NLP models using HuggingFace Transformers for real-time sentiment classification and context-aware response generation

MACHINE LEARNING & AI PROJECTS

Institutional-Grade Algorithmic Trading Engine | *Python, FastAPI, Streamlit, HMM, Alpaca API* GitHub

- Engineered production-grade quantitative trading system with Hidden Markov Model market regime detection, Black-Litterman portfolio optimization, and CVaR risk management – equivalent in architecture to institutional trading desks
- Built full-stack application with FastAPI backend and Streamlit dashboard, integrated with Alpaca paper trading API for live order execution and real-time portfolio monitoring
- Implemented dynamic position sizing, automated rebalancing, and multi-asset allocation across NSE and NYSE markets with risk alerts – demonstrating end-to-end ML model deployment in a live environment

Stock Price Prediction System | *Scikit-learn, Streamlit, FastAPI, Python* Live Demo | GitHub

- Deployed ML forecasting system serving 500+ Indian and US stocks with 85% directional accuracy using 15+ engineered features including RSI, MACD, Bollinger Bands, and support/resistance levels
- Built scalable inference pipeline with FastAPI model serving, prediction caching, and interactive Plotly/Streamlit dashboard with real-time candlestick visualization

Intelligent Emotional Support Chatbot | *Python, HuggingFace, PyTorch, Flask, React*

- Built NLP-powered chatbot combining a PyTorch intent classifier with HuggingFace transformer models for real-time sentiment analysis and context-aware emotional support responses
- Deployed full-stack application with React frontend and Flask backend, featuring session memory and multi-turn conversation handling – demonstrating practical LLM integration and deployment

E-Commerce Sales Analytics & Predictive Pipeline | *Python, SQL, AWS, ML* GitHub

- Built end-to-end ETL pipeline processing 3,497 transactions on AWS RDS with automated customer segmentation, churn prediction, and cohort analysis using advanced SQL window functions and CTEs
- Identified 12.4% return rate anomaly and seasonal revenue patterns through hypothesis testing, delivering actionable business insights that reduced data processing time by 40%

Electoral Data Anomaly Detection (EVOD) | *Python, Pandas, Streamlit* Live Demo | GitHub

- Developed statistical anomaly detection system for 100K+ records using Z-score, IQR, and distribution fitting achieving 95% precision with automated data quality reporting and stakeholder dashboard

Constraint Optimization Scheduler (OR-Tools) | *Python, Django, Google OR-Tools*

- Implemented constraint satisfaction algorithm handling 200+ faculty and 3,000+ students, solving complex scheduling across 50+ classes with conflict minimization and automated PDF/CSV reporting

CERTIFICATIONS

- IBM Data Science Professional Certificate (9-course) – ML, Python, SQL, Statistical Analysis
- Google Cloud Data Analytics – BigQuery, Cloud-based ML, Scalable Analytics Pipelines
- IBM Data Engineering Fundamentals – ETL Pipelines, Feature Engineering, Data Architecture
- Microsoft Azure Fundamentals (AZ-900) – Cloud ML Services and Azure Machine Learning
- NPTEL SWAYAM – Data Analysis and Decision Making (IIT Kanpur)

ADDITIONAL INFORMATION

Languages: English (Fluent), Malayalam (Native), Tamil (Fluent), Hindi (Fluent), Arabic (Basic)

Interests: Quantitative Finance, Algorithmic Trading, Financial ML, NLP Applications, Generative AI, LLMOps